Constant rates of change

1

a Constant **b** Variable **c** Constant **d** Variable

e Variable **f** Variable

2 Constant

Average rates of change

3

a -4 **b** -3 **c** -7 **d** Variable

4

a -1 **b** -1 **c** -1 **d** Linear

5 Non-linear

6

a -4 **b** -4 **c** -4 **d** Linear

7 $-\frac{30}{\pi}$

8

a 1

b 2

c 4

d

Interval	$[3,4]$	$[4,5]$	$[5,6]$
Average rate of change	8	16	32

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Instantaneous rates of change

9 0

10
a 0 **b** 0

11 2

12
a 2 **b** 2

13
a 0 **b** 0 **c** 2 **d** 1
e -2 **f** 2 **g** -1 **h** 1
i -3

14
a 10 **b** 7

15
a -4 **b** -6.5

16
a 1 **b** 2

17
a -3 **b** -2

18
a $\frac{2}{3}$ **b** $\frac{1}{2}$

19
a 6 **b** 2

20
a -3 **b** -2

21

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a $a = 7$ **b** $b = 5$ **c** 2

22

a

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
1	2	1	7
1	1.5	0.5	4.75
1	1.1	0.1	3.31
1	1.05	0.05	3.1525
1	1.01	0.01	3.0301
1	1.001	0.001	3.003
1	1.0001	0.0001	3.0003

b 3

23

a

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
1	2	1	6
1	1.5	0.5	5
1	1.1	0.1	4.2
1	1.05	0.05	4.1
1	1.01	0.01	4.02
1	1.001	0.001	4.002
1	1.0001	0.0001	4.0002

b 4

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24

a

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
0	1	1	3
0	0.5	0.5	2
0	0.1	0.1	1.487
0	0.05	0.05	1.4355
0	0.01	0.01	1.3959
0	0.001	0.001	$0.138\,73$
0	0.0001	0.0001	1.3864

b 1.386

25

a

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
1	2	1	-3
1	1.5	0.5	-2.5
1	1.1	0.1	-2.1
1	1.05	0.05	-2.05
1	1.01	0.01	-2.01
1	1.001	0.001	-2.001
1	1.0001	0.0001	-2.0001

b -2

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26

a

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
1	2	1	-2.5
1	1.5	0.5	-3.3333
1	1.1	0.1	-4.5455
1	1.05	0.05	-4.7619
1	1.01	0.01	-4.9505
1	1.001	0.001	-4.995
1	1.0001	0.0001	-4.9995

b -5

Applications

27

a 11 **b** 26 **c** $\frac{15}{4}$

28

a 0 **b** 3536 **c** 272

29 $\$300$ per year

30 60

31

a $y = 4x + 20$ **b** $y = 20\,(1.10)^x$

c 4 turbines each year **d** 9.59 turbines each year

e Option A

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32

a

i $\$4$ per year **ii** $\$2$ per year **iii** $\$6$ per year **iv** $\$1$ per year

b Between 1996 and 2010

33

a

i $-\$5$ per year **ii** $-\$6$ per year **iii** $-\$3$ per year **iv** $-\$1$ per year

b Between 1985 and 1999

34

a

i $\$4.30$ per year **ii** $\$0.20$ per year

iii $\$6.10$ per year **iv** $-\$1.40$ per year

b Between years 13 and 26.

35

a

i 19 km/h **ii** 18 km/h **iii** 11 km/h **iv** 15 km/h

b The first hour.

36 $-\$10$ per $1^\circ F$

37

a

Year	0	1	2	3	4	5
Tobia's staff	310	269	228	187	146	105
Gwen's staff	310	302	286	262	230	190

b -41

c -32

d Linear, because it decreases by the same amount each year.

e Non-linear, because it decreases by more each year.

f Tobias's

g Gwen's

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38

a

Month	0	1	2	3	4	5
Mario's clients	0	27	54	81	108	135
Christa's clients	0	6	18	36	60	90

b Linear

c Non-linear

d 27

e 18

f Mario's

f Christa's

39

a

Month	0	1	2	3	4	5
Value of Harry's investment	2000	2060	2120	2180	2240	2300
Value of Skye's investment	2000	2040	2080.8	2122.42	2164.87	2208.17

b $\$60$

c $\$40.8$

d Linear

e Non-linear

f Harry's

40

a $-\frac{1}{2}$

b The amount of water in the bucket decreases by $\frac{1}{2}$ litre every minute.

41

a $\frac{3}{5}$

b The cost of a call increases by $\$0.60$ for each additional minute

42 Oliver, $10\ \text{km/h}$

43 6.5 insects per week

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44

a 0°F per hour **b** 6°F per hour **c** -1°F per hour **d** 1.5°F per hour

45

a

i $9\ \text{m/s}$ **ii** $3\ \text{m/s}$ **iii** $-3\ \text{m/s}$ **iv** $-9\ \text{m/s}$

b Between $t = 0$ and $t = 1$, and between $t = 3$ and $t = 4$

f The ball is falling down.

46

a 130 **b** 114 **c** Product A

47

a 1.20 **b** 0.37 **c** Fund A

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